Practical -01

Aim:- Develop a course registration application where students can register courses.

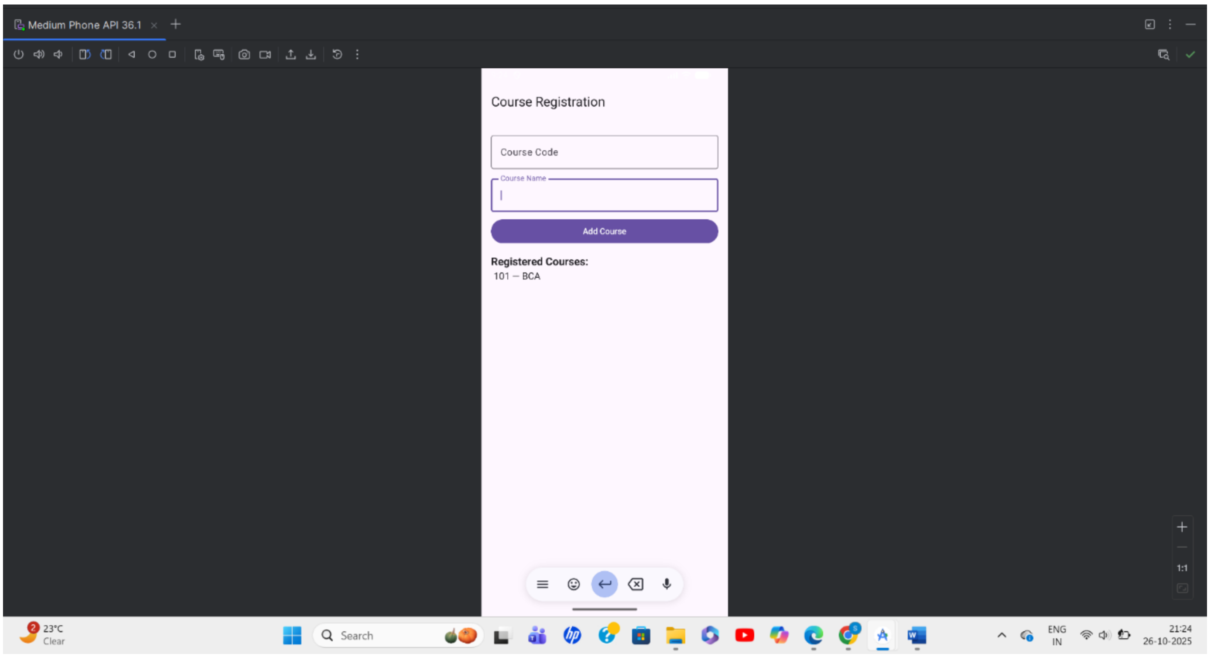
Source code :

```
package com.example.courseregistration

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.lazy.LazyColumn
import androidx.compose.foundation.lazy.items
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Modifier
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.tooling.preview.Preview
import androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContent {
            CourseRegistrationApp()
        }
    }
}

@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun CourseRegistrationApp() {
    var courseCode by remember { mutableStateOf("") }
    var courseName by remember { mutableStateOf("") }
    var courses by remember { mutableStateOf(listOf<Pair<String, String>>()) }

    Scaffold(
        topBar = {
            TopAppBar(title = { Text("Course Registration") })
        }
    ) { padding ->
        Column(
            modifier = Modifier
                .padding(padding)
```

```
            .padding(16.dp)
    ) {
        OutlinedTextField(
            value = courseCode,
            onValueChange = { courseCode = it },
            label = { Text("Course Code") },
            modifier = Modifier.fillMaxWidth()
        )

        Spacer(modifier = Modifier.height(8.dp))

        OutlinedTextField(
            value = courseName,
            onValueChange = { courseName = it },
            label = { Text("Course Name") },
            modifier = Modifier.fillMaxWidth()
        )

        Spacer(modifier = Modifier.height(8.dp))

        Button(
            onClick = {
                if (courseCode.isNotBlank() && courseName.isNotBlank()) {
                    courses = courses + (courseCode to courseName)
                    courseCode = ""
                    courseName = ""
                }
            },
            modifier = Modifier.fillMaxWidth()
        ) {
            Text("Add Course")
        }

        Spacer(modifier = Modifier.height(16.dp))

        Text(
            text = "Registered Courses:",
            fontSize = 18.sp,
            fontWeight = FontWeight.Bold
        )

        LazyColumn {
            items(courses) { (code, name) ->
                Text(
                    text = "$code — $name",
                    fontSize = 16.sp,
```

```
                modifier = Modifier.padding(4.dp)
            )
        }
    }
  }
 }
}

@Preview(showBackground = true)
@Composable
fun CourseRegistrationAppPreview() {
    CourseRegistrationApp()
}
```

Output:



Practical -02

Aim:- Develop a college fee management application to get fee details of students.

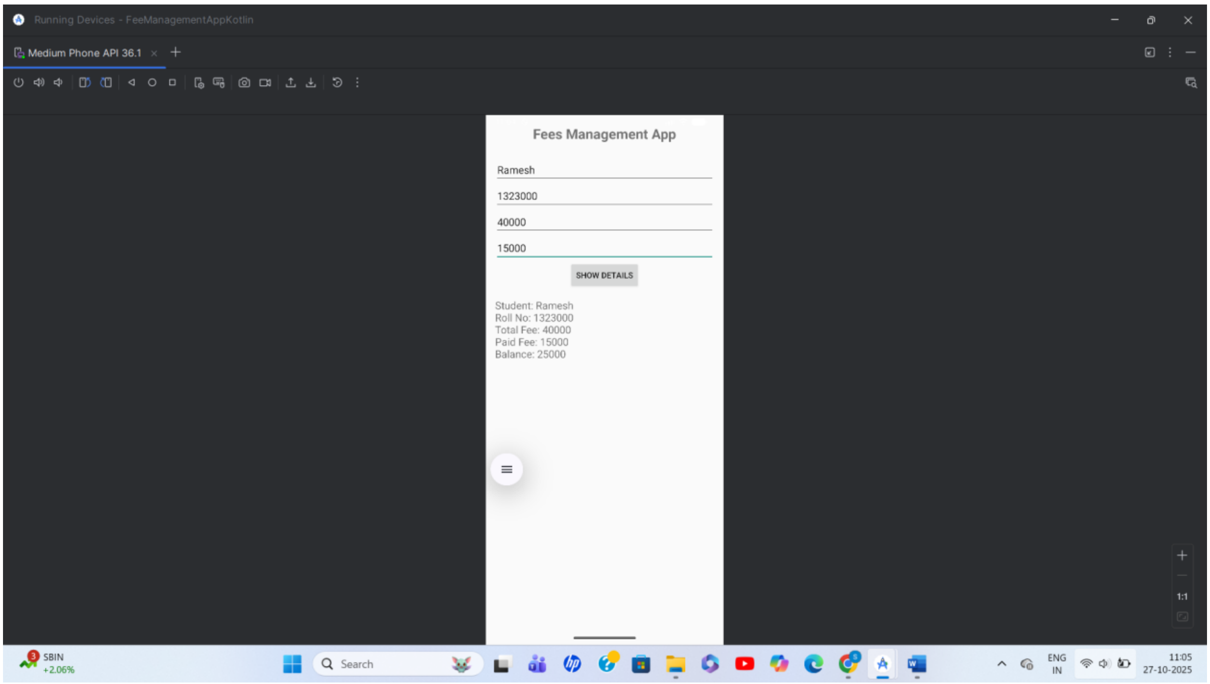
```
package com.example.feemanagementappkotlin

import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.*

class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val nameEdit = findViewById<EditText>(R.id.nameEdit)
        val rollEdit = findViewById<EditText>(R.id.rollEdit)
        val totalFeeEdit = findViewById<EditText>(R.id.totalFeeEdit)
        val paidFeeEdit = findViewById<EditText>(R.id.paidFeeEdit)
        val btnShow = findViewById<Button>(R.id.btnShow)
        val resultText = findViewById<TextView>(R.id.resultText)

        btnShow.setOnClickListener {
            val name = nameEdit.text.toString()
            val roll = rollEdit.text.toString()
            val totalFee = totalFeeEdit.text.toString()
            val paidFee = paidFeeEdit.text.toString()

            if (name.isEmpty() || roll.isEmpty() || totalFee.isEmpty() || paidFee.isEmpty()) {
                resultText.text = "Please fill all fields!"
            } else {
                val total = totalFee.toInt()
                val paid = paidFee.toInt()
                val balance = total - paid

                resultText.text = """
                    Student: $name
                    Roll No: $roll
                    Total Fee: $total
                    Paid Fee: $paid
                    Balance: $balance
                """.trimIndent()
            }
        }
    }
}
```

//activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Fees Management App"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_gravity="center_horizontal"
        android:layout_marginBottom="24dp"/>

    <EditText
        android:id="@+id/nameEdit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Name" />

    <EditText
        android:id="@+id/rollEdit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Roll No"
        android:inputType="number" />

    <EditText
        android:id="@+id/totalFeeEdit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Total Fee"
        android:inputType="numberDecimal" />

    <EditText
        android:id="@+id/paidFeeEdit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Paid Fee"
        android:inputType="numberDecimal" />
```

```xml
    <Button
        android:id="@+id/btnShow"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center_horizontal"
        android:text="Show Details" />

    <TextView
        android:id="@+id/resultText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:textSize="18sp"
        tools:text="Result will be displayed here" />

</LinearLayout>
```

Output:



Practical -03

Aim:- Create an Android app using Spinner and Date Picker to collect user data.

```
package com.example.spinnerdatepickerapp

import android.app.DatePickerDialog
import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity
import java.util.*

class MainActivity : AppCompatActivity() {

    var selectedCourse = ""
    var selectedDate = ""

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val nameEdit = findViewById<EditText>(R.id.nameEdit)
        val spinner = findViewById<Spinner>(R.id.spinner)
        val dateBtn = findViewById<Button>(R.id.dateBtn)
        val dateText = findViewById<TextView>(R.id.dateText)
        val submitBtn = findViewById<Button>(R.id.submitBtn)
        val resultText = findViewById<TextView>(R.id.resultText)

        // Setup Spinner
        ArrayAdapter.createFromResource(
            this,
            R.array.courses,
            android.R.layout.simple_spinner_item
        ).also { adapter ->
adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item)
            spinner.adapter = adapter
        }
        spinner.onItemSelectedListener = object : AdapterView.OnItemSelectedListener {
            override fun onItemSelected(parent: AdapterView<*>, view: android.view.View,
position: Int, id: Long) {
                selectedCourse = parent.getItemAtPosition(position).toString()
            }
            override fun onNothingSelected(parent: AdapterView<*>) {}
        }
```

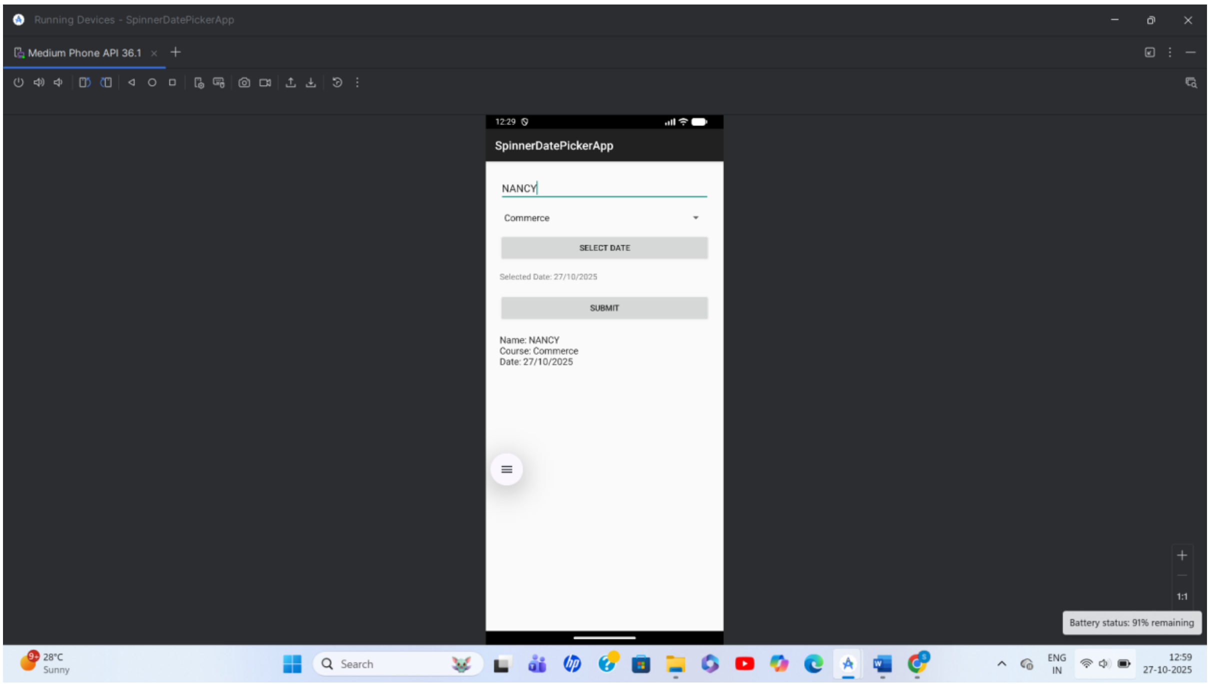
```
        // Date Picker Dialog
        dateBtn.setOnClickListener {
            val c = Calendar.getInstance()
            val year = c.get(Calendar.YEAR)
            val month = c.get(Calendar.MONTH)
            val day = c.get(Calendar.DAY_OF_MONTH)

            val datePickerDialog = DatePickerDialog(
                this,
                { _, selYear, selMonth, selDay ->
                    selectedDate = "$selDay/${selMonth + 1}/$selYear"
                    dateText.text = "Selected Date: $selectedDate"
                },
                year,
                month,
                day
            )
            datePickerDialog.show()
        }

        // Submit Button
        submitBtn.setOnClickListener {
            val name = nameEdit.text.toString()
            if (name.isEmpty() || selectedCourse.isEmpty() || selectedDate.isEmpty()) {
                resultText.text = "Please fill all fields and select a date."
            } else {
                resultText.text = """
                    Name: $name
                    Course: $selectedCourse
                    Date: $selectedDate
                """.trimIndent()
            }
        }
    }
}
```

// activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp">

    <EditText
        android:id="@+id/nameEdit"
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:hint="Enter your name" />

    <Spinner
        android:id="@+id/spinner"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp" />

    <Button
        android:id="@+id/dateBtn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="Select Date" />

    <TextView
        android:id="@+id/dateText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="Selected date will appear here" />

    <Button
        android:id="@+id/submitBtn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Submit" />

    <TextView
        android:id="@+id/resultText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:textColor="#272727"
        android:textSize="16sp" />
</LinearLayout>
```

Output:


Running Devices - SpinnerDatePickerApp
Medium Phone API 36.1
SpinnerDatePickerApp
NANCY
Commerce
SELECT DATE
Selected Date: 27/10/2025
SUBMIT
Name: NANCY
Course: Commerce
Date: 27/10/2025
Battery status: 91% remaining
28°C
Sunny
Search
12:59
27-10-2025

Practical -04

Aim:- Develop a currency converter application to help users.

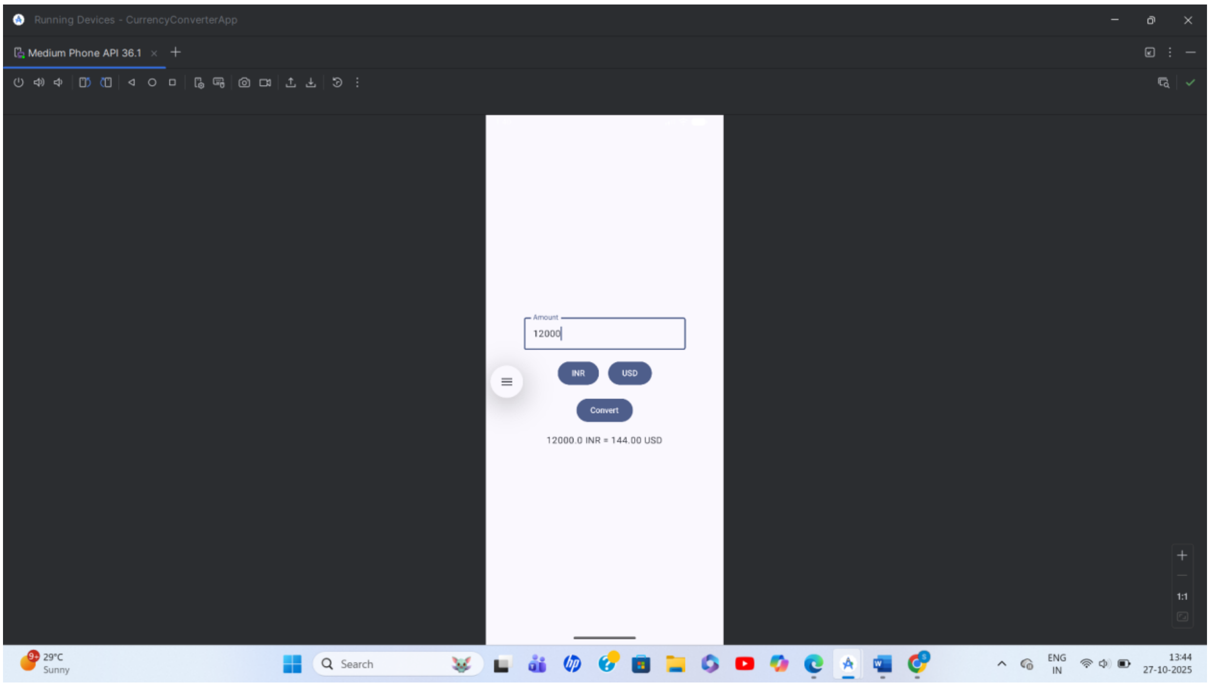
```
package com.example.currencyconverterapp

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.text.KeyboardOptions
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.text.input.KeyboardType
import androidx.compose.ui.tooling.preview.Preview
import androidx.compose.ui.unit.dp
import com.example.currencyconverterapp.ui.theme.CurrencyConverterAppTheme

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContent {
            CurrencyConverterAppTheme {
                Surface(
                    modifier = Modifier.fillMaxSize(),
                    color = MaterialTheme.colorScheme.background
                ) {
                    CurrencyConverter()
                }
            }
        }
    }
}

@Composable
fun CurrencyConverter() {
    var amount by remember { mutableStateOf("") }
    var fromCurrency by remember { mutableStateOf("USD") }
    var toCurrency by remember { mutableStateOf("INR") }
    var result by remember { mutableStateOf("") }
    val currencies = listOf("USD", "INR", "EUR")

    // Hardcoded rates, example: USD to INR/EUR, INR to USD/EUR, EUR to USD/INR
    fun getRate(from: String, to: String): Double {
        return when (from) {
```

```
        "USD" -> when (to) {
            "INR" -> 83.0
            "EUR" -> 0.93
            "USD" -> 1.0
            else -> 1.0
        }
        "INR" -> when (to) {
            "USD" -> 0.012
            "EUR" -> 0.011
            "INR" -> 1.0
            else -> 1.0
        }
        "EUR" -> when (to) {
            "USD" -> 1.08
            "INR" -> 89.0
            "EUR" -> 1.0
            else -> 1.0
        }
        else -> 1.0
    }
}

Column(
    modifier = Modifier
        .fillMaxSize()
        .padding(16.dp),
    horizontalAlignment = Alignment.CenterHorizontally,
    verticalArrangement = Arrangement.Center
) {
    OutlinedTextField(
        value = amount,
        onValueChange = { amount = it },
        label = { Text("Amount") },
        keyboardOptions = KeyboardOptions(keyboardType = KeyboardType.Number)
    )
    Spacer(modifier = Modifier.height(16.dp))
    Row {
        Spinner(
            currencies = currencies,
            selectedCurrency = fromCurrency,
            onCurrencySelected = { fromCurrency = it }
        )
        Spacer(modifier = Modifier.width(16.dp))
        Spinner(
            currencies = currencies,
            selectedCurrency = toCurrency,
```

```
                onCurrencySelected = { toCurrency = it }
            )
        }
        Spacer(modifier = Modifier.height(16.dp))
        Button(onClick = {
            if (amount.isEmpty()) {
                result = "Please enter amount!"
            } else {
                try {
                    val amountValue = amount.toDouble()
                    val rate = getRate(fromCurrency, toCurrency)
                    val converted = amountValue * rate
                    result = "$amountValue $fromCurrency = %.2f $toCurrency".format(converted)
                } catch (e: NumberFormatException) {
                    result = "Invalid amount"
                }
            }
        }) {
            Text("Convert")
        }
        Spacer(modifier = Modifier.height(16.dp))
        Text(result)
    }
}

@Composable
fun Spinner(
    currencies: List<String>,
    selectedCurrency: String,
    onCurrencySelected: (String) -> Unit
) {
    var expanded by remember { mutableStateOf(false) }

    Box {
        Button(onClick = { expanded = true }) {
            Text(selectedCurrency)
        }
        DropdownMenu(
            expanded = expanded,
            onDismissRequest = { expanded = false }
        ) {
            currencies.forEach { currency ->
                DropdownMenuItem(
                    text = { Text(currency) },
                    onClick = {
                        onCurrencySelected(currency)
```

```
                        expanded = false
                    }
                )
            }
        }
    }
}

@Preview(showBackground = true)
@Composable
fun DefaultPreview() {
    CurrencyConverterAppTheme {
        CurrencyConverter()
    }
}
```

Output:



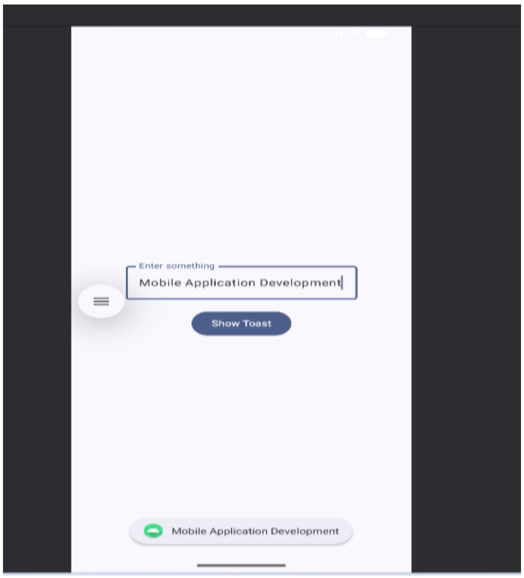
Practical -05

Aim:- Design an app using multiple widgets: Button, Toast, and Text Fields.

```
package com.example.texttoastdemo

import android.os.Bundle
import android.widget.Toast
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.Arrangement
import androidx.compose.foundation.layout.Column
import androidx.compose.foundation.layout.fillMaxSize
import androidx.compose.foundation.layout.padding
import androidx.compose.material3.Button
import androidx.compose.material3.OutlinedTextField
import androidx.compose.material3.Surface
import androidx.compose.material3.Text
import androidx.compose.runtime.Composable
import androidx.compose.runtime.getValue
import androidx.compose.runtime.mutableStateOf
import androidx.compose.runtime.remember
import androidx.compose.runtime.setValue
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.platform.LocalContext
import androidx.compose.ui.tooling.preview.Preview
import androidx.compose.ui.unit.dp
import com.example.texttoastdemo.ui.theme.TextToastDemoTheme

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContent {
            TextToastDemoTheme {
                Surface(modifier = Modifier.fillMaxSize()) {
                    ToastScreen()
                }
            }
        }
    }
}
@Composable
fun ToastScreen() {
    var textState by remember { mutableStateOf("") }
    val context = LocalContext.current
```

```
    Column(
        modifier = Modifier
            .fillMaxSize()
            .padding(16.dp),
        horizontalAlignment = Alignment.CenterHorizontally,
        verticalArrangement = Arrangement.Center
    ) {
        OutlinedTextField(
            value = textState,
            onValueChange = { textState = it },
            label = { Text("Enter something") }
        )
        Button(
            onClick = {
                Toast.makeText(context, textState, Toast.LENGTH_LONG).show()
            },
            modifier = Modifier.padding(top = 16.dp)
        ) {
            Text("Show Toast")
        }
    }
}

@Preview(showBackground = true)
@Composable
fun DefaultPreview() {
    TextToastDemoTheme {
        ToastScreen()
    }
}
```

Output:



Practical -06

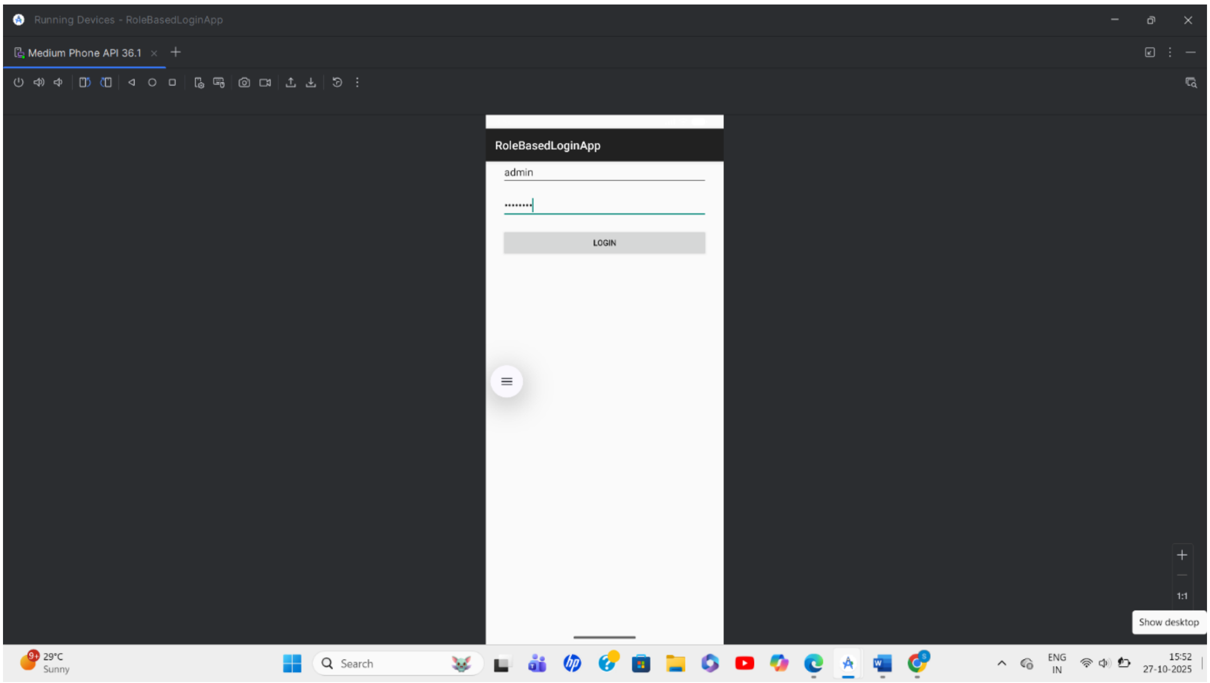
Aim:- Develop a login system with role-based applications like admin & user.

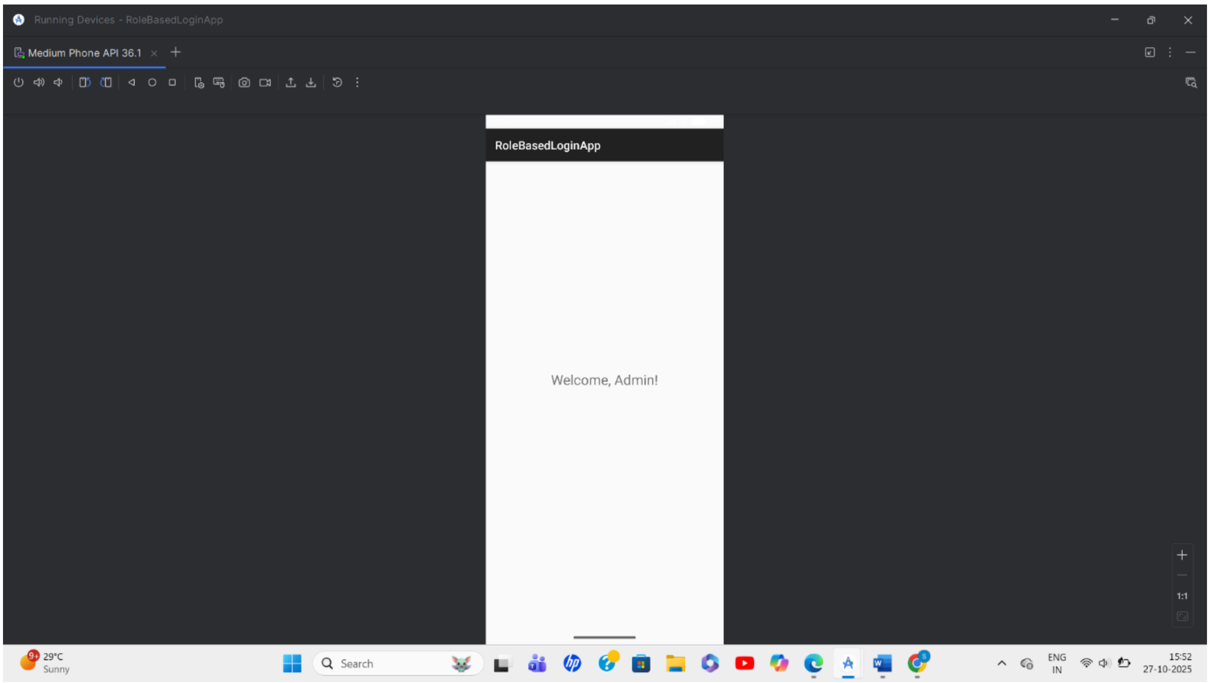
```
package com.example.rolebasedloginapp

import android.content.Intent
import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val usernameEdit = findViewById<EditText>(R.id.usernameEdit)
        val passwordEdit = findViewById<EditText>(R.id.passwordEdit)
        val loginBtn = findViewById<Button>(R.id.loginBtn)
        val errorText = findViewById<TextView>(R.id.errorText)

        loginBtn.setOnClickListener {
            val username = usernameEdit.text.toString().trim()
            val password = passwordEdit.text.toString().trim()
            // Hardcoded users
            if(username == "admin" && password == "admin123") {
                startActivity(Intent(this, AdminActivity::class.java))
                finish()
            } else if(username == "user" && password == "user123") {
                startActivity(Intent(this, UserActivity::class.java))
                finish()
            } else {
                errorText.text = "Invalid credentials. Try again!"
            }
        }
    }
}
```

```
//activity_main.xml

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="28dp"
    android:orientation="vertical">
```

```
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login"
        android:textSize="24sp"
        android:gravity="center"
        android:paddingBottom="16dp" />

    <EditText
        android:id="@+id/usernameEdit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username"/>

    <EditText
        android:id="@+id/passwordEdit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:inputType="textPassword"
        android:hint="Password"
        android:layout_marginTop="12dp"/>

    <Button
        android:id="@+id/loginBtn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login"
        android:layout_marginTop="18dp"/>

    <TextView
        android:id="@+id/errorText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textColor="#D32F2F"
        android:layout_marginTop="16dp"/>
</LinearLayout>
```

Output:


Running Devices - RoleBasedLoginApp
Medium Phone API 36.1
RoleBasedLoginApp
admin
LOGIN
Show desktop
29°C
Sunny
Search
ENG
IN
15:52
27-10-2025


Running Devices - RoleBasedLoginApp
Medium Phone API 36.1
RoleBasedLoginApp
Welcome, Admin!
29°C
Sunny
Search
ENG
IN
15:52
27-10-2025

Practical -07

Aim:- Develop a bus management application to check student bus seat allocation.

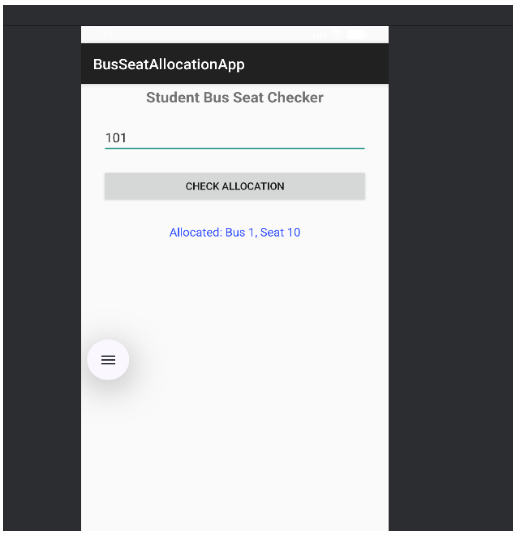
```
package com.example.busseatallocationapp

import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.*

class MainActivity : AppCompatActivity() {
    // Example student allocation data: Roll to seat mapping
    private val allocations = mapOf(
        "101" to "Bus 1, Seat 10",
        "102" to "Bus 1, Seat 11",
        "103" to "Bus 2, Seat 03",
        "104" to "Bus 3, Seat 07"
    )
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val rollEdit = findViewById<EditText>(R.id.rollEdit)
        val checkBtn = findViewById<Button>(R.id.checkBtn)
        val resultText = findViewById<TextView>(R.id.resultText)

        checkBtn.setOnClickListener {
            val roll = rollEdit.text.toString().trim()
            if (roll.isEmpty()) {
                resultText.text = "Please enter a roll number."
            } else if (allocations.containsKey(roll)) {
                resultText.text = "Allocated: ${allocations[roll]}"
            } else {
                resultText.text = "No seat allocation found for roll number: $roll"
            }
        }
    }
}
```

```
//activity_main.xml
```

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="28dp">
```

```
  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="center"
    android:paddingBottom="20dp"
    android:text="Student Bus Seat Checker"
    android:textSize="20sp"
    android:textStyle="bold" />

  <EditText
    android:id="@+id/rollEdit"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Roll Number" />

  <Button
    android:id="@+id/checkBtn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:text="Check Allocation" />

  <TextView
    android:id="@+id/resultText"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="28dp"
    android:gravity="center"
    android:textColor="#304FFE"
    android:textSize="16sp" />
</LinearLayout>
```

Output:



Practical -08

Aim:- Develop a university event management application to track events.

```
package com.example.universityeventmanager

import android.app.DatePickerDialog
import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity
import java.util.*

class MainActivity : AppCompatActivity() {

    private lateinit var eventTitle: EditText
    private lateinit var eventDescription: EditText
    private lateinit var eventDate: EditText
    private lateinit var addEventBtn: Button
    private lateinit var eventsList: ListView

    private val eventItems = mutableListOf<String>()
    private lateinit var adapter: ArrayAdapter<String>

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        eventTitle = findViewById(R.id.eventTitle)
        eventDescription = findViewById(R.id.eventDescription)
        eventDate = findViewById(R.id.eventDate)
        addEventBtn = findViewById(R.id.addEventBtn)
        eventsList = findViewById(R.id.eventsList)

        adapter = ArrayAdapter(this, android.R.layout.simple_list_item_1, eventItems)
        eventsList.adapter = adapter

        eventDate.setOnClickListener {
            val calendar = Calendar.getInstance()
            val dpd = DatePickerDialog(
                this,
                { _, year, month, dayOfMonth ->
                    eventDate.setText("$dayOfMonth/${month + 1}/$year")
                },
                calendar.get(Calendar.YEAR),
                calendar.get(Calendar.MONTH),
                calendar.get(Calendar.DAY_OF_MONTH)
            )
```

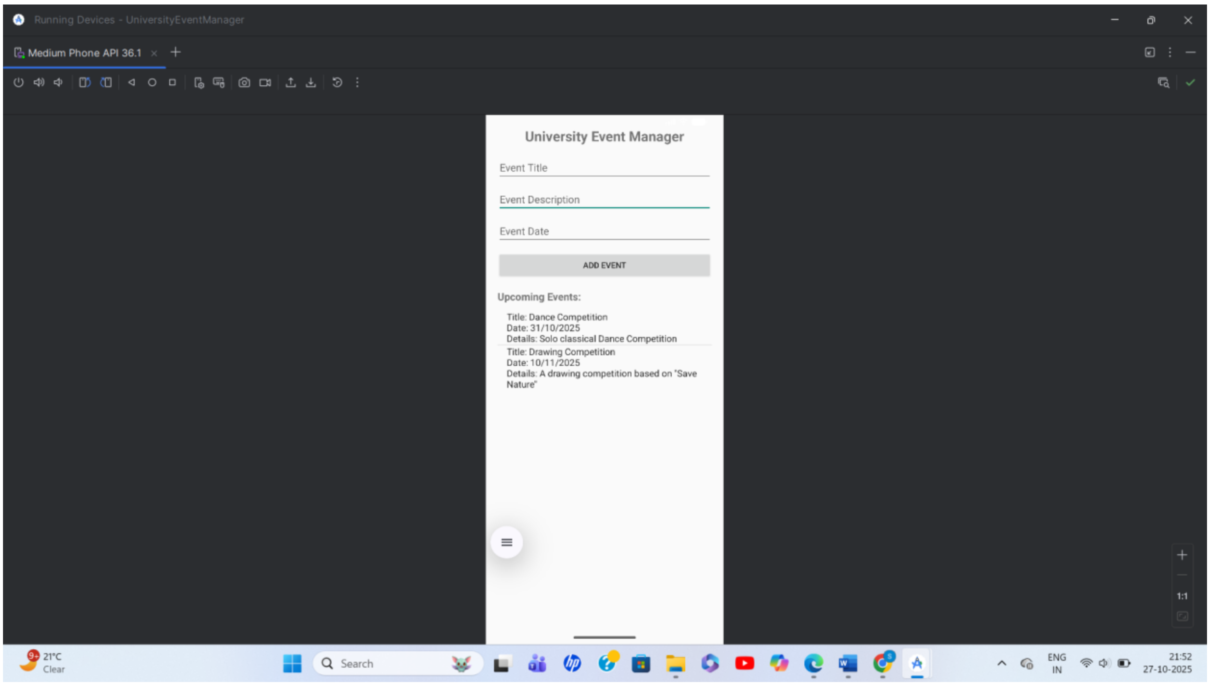
```
            dpd.show()
        }

        addEventBtn.setOnClickListener {
            val title = eventTitle.text.toString().trim()
            val desc = eventDescription.text.toString().trim()
            val date = eventDate.text.toString().trim()

            if (title.isEmpty() || desc.isEmpty() || date.isEmpty()) {
                Toast.makeText(this, "Please fill all event details", Toast.LENGTH_SHORT).show()
            } else {
                val eventInfo = "Title: $title\nDate: $date\nDetails: $desc"
                eventItems.add(eventInfo)
                adapter.notifyDataSetChanged()

                eventTitle.text.clear()
                eventDescription.text.clear()
                eventDate.text.clear()
                Toast.makeText(this, "Event added successfully", Toast.LENGTH_SHORT).show()
            }
        }
    }
}
```

//activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="20dp">

    <TextView
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:text="University Event Manager" android:textSize="24sp"
        android:textStyle="bold" android:layout_gravity="center"
        android:paddingBottom="16dp"/>

    <EditText
        android:id="@+id/eventTitle"
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:hint="Event Title" />

    <EditText
        android:id="@+id/eventDescription"
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:hint="Event Description" android:layout_marginTop="10dp" />

    <EditText
```

```
        android:id="@+id/eventDate"
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:hint="Event Date" android:focusable="false"
android:layout_marginTop="10dp" />

    <Button
        android:id="@+id/addEventBtn"
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:text="Add Event" android:layout_marginTop="12dp"/>

    <TextView
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:text="Upcoming Events:" android:textSize="18sp"
        android:textStyle="bold" android:layout_marginTop="18dp"/>

    <ListView
        android:id="@+id/eventsList"
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:layout_marginTop="12dp" />
</LinearLayout>
```

Output:



Practical -09

Aim:- Build an Academic Calendar Manager application to get details.

```
package com.example.academiccalendarmanager

import android.app.AlertDialog
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity
import androidx.core.content.ContextCompat
import com.applandeo.materialcalendarview.CalendarView
import com.applandeo.materialcalendarview.listeners.OnCalendarDayClickListener
import java.text.SimpleDateFormat
import java.util.*

class MainActivity : AppCompatActivity() {

    // In-memory mapping of dates to event description
    private val eventsMap = mutableMapOf<String, String>()
    private val calendarDays =
mutableListOf<com.applandeo.materialcalendarview.CalendarDay>()

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val calendarView = findViewById<CalendarView>(R.id.calendarView)
        val eventDetailsView = findViewById<TextView>(R.id.eventDetailsView)
        val addEventButton = findViewById<Button>(R.id.addEventButton)

        fun dateKey(calendar: Calendar) = "${calendar.get(Calendar.YEAR)}-
${calendar.get(Calendar.MONTH)}-${calendar.get(Calendar.DAY_OF_MONTH)}"

        val dateFormat = SimpleDateFormat("dd MMMM yyyy", Locale.getDefault())

        calendarView.setOnCalendarDayClickListener(object : OnCalendarDayClickListener {
            override fun onClick(calendarDay:
com.applandeo.materialcalendarview.CalendarDay) {
                val calendar = calendarDay.calendar
                val key = dateKey(calendar)
                val info = eventsMap[key]
                if (info != null) {
                    eventDetailsView.text = "Event on ${dateFormat.format(calendar.time)}:\n$info"
                } else {
```

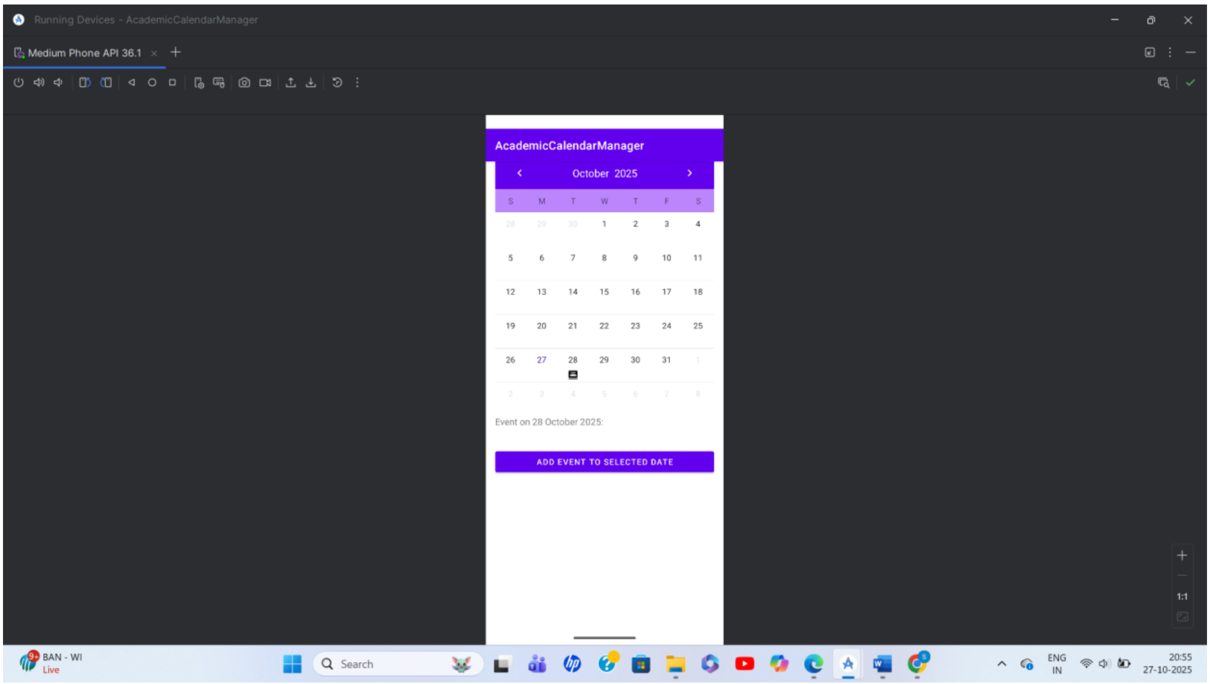
```
                eventDetailsView.text = "No event for this date."
            }
        }
    })

    addEventButton.setOnClickListener {
        val selectedCalendar = calendarView.firstSelectedDate ?: Calendar.getInstance()
        val input = EditText(this)
        AlertDialog.Builder(this)
            .setTitle("Add Academic Event")
            .setMessage("Enter event details for
${dateFormat.format(selectedCalendar.time)}:")
            .setView(input)
            .setPositiveButton("Save") { _, _ ->
                val key = dateKey(selectedCalendar)
                eventsMap[key] = input.text.toString()

                val eventIcon = ContextCompat.getDrawable(this, R.drawable.ic_event_note)
                val calendarDay =
com.applandeo.materialcalendarview.CalendarDay(selectedCalendar)
                calendarDay.imageDrawable = eventIcon
                calendarDays.add(calendarDay)

                calendarView.setCalendarDays(calendarDays)
                eventDetailsView.text = "Event Saved!"
            }
            .setNegativeButton("Cancel", null)
            .show()
    }
  }
}
```

//activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".MainActivity">

    <com.applandeo.materialcalendarview.CalendarView
        android:id="@+id/calendarView"
        android:layout_width="match_parent"
```

```xml
        android:layout_height="430dp"
        app:abbreviationsBarColor="@color/purple_200"
        app:headerColor="@color/purple_500"
        app:pagesColor="@android:color/white"
        app:selectionColor="@color/purple_200"
        app:selectionLabelColor="@android:color/white"
        app:todayLabelColor="@color/purple_700" />

    <TextView
        android:id="@+id/eventDetailsView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="Select a date to see event details"
        android:textSize="16sp" />

    <Button
        android:id="@+id/addEventButton"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="Add Event to Selected Date" />

</LinearLayout>
```

Output:



Practical -10

Aim:- Develop a Library Book Management application to track issued books

```
package com.example.librarybookmanagement

import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    private lateinit var bookNameEdit: EditText
    private lateinit var studentNameEdit: EditText
    private lateinit var issueBookBtn: Button
    private lateinit var returnBookBtn: Button
    private lateinit var statusTextView: TextView

    // In-memory store of issued books: book name -> student name
    private val issuedBooksMap = mutableMapOf<String, String>()

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        bookNameEdit = findViewById(R.id.bookNameEdit)
        studentNameEdit = findViewById(R.id.studentNameEdit)
        issueBookBtn = findViewById(R.id.issueBookBtn)
        returnBookBtn = findViewById(R.id.returnBookBtn)
        statusTextView = findViewById(R.id.statusTextView)

        updateStatus()

        issueBookBtn.setOnClickListener {
            val bookName = bookNameEdit.text.toString().trim()
            val studentName = studentNameEdit.text.toString().trim()

            when {
                bookName.isEmpty() -> showToast("Enter Book Name")
                studentName.isEmpty() -> showToast("Enter Student Name")
                issuedBooksMap.containsKey(bookName) -> showToast("Book is already
issued!")
                else -> {
                    issuedBooksMap[bookName] = studentName
                    showToast("Book issued successfully")
                    clearFields()
```

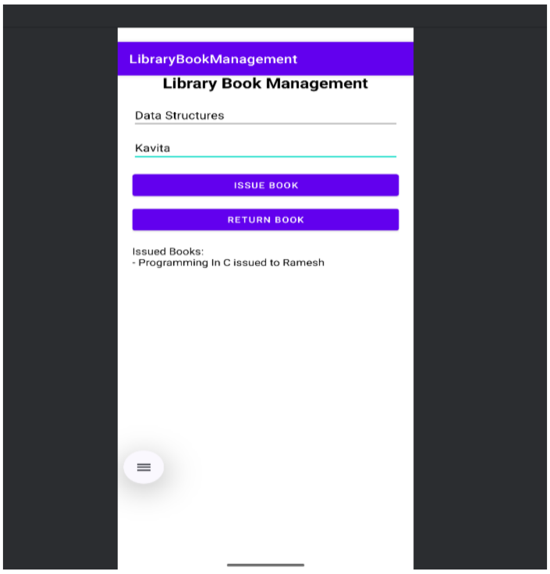
```
                updateStatus()
            }
        }
    }

    returnBookBtn.setOnClickListener {
        val bookName = bookNameEdit.text.toString().trim()
        if (issuedBooksMap.containsKey(bookName)) {
            issuedBooksMap.remove(bookName)
            showToast("Book returned successfully")
            clearFields()
            updateStatus()
        } else {
            showToast("This book was not issued.")
        }
    }
}

private fun updateStatus() {
    if (issuedBooksMap.isEmpty()) {
        statusTextView.text = "No books issued currently."
    } else {
        val status = StringBuilder("Issued Books:\n")
        issuedBooksMap.forEach { (book, student) ->
            status.append("- $book issued to $student\n")
        }
        statusTextView.text = status.toString()
    }
}

private fun clearFields() {
    bookNameEdit.text.clear()
    studentNameEdit.text.clear()
}

private fun showToast(message: String) {
    Toast.makeText(this, message, Toast.LENGTH_SHORT).show()
}
}
```

//activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="20dp">

    <TextView
```

```
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:text="Library Book Management" android:textSize="24sp"
        android:textStyle="bold" android:layout_gravity="center"
        android:textColor="@android:color/black"
        android:paddingBottom="16dp"/>

    <EditText
        android:id="@+id/bookNameEdit"
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:hint="Book Name"
        android:textColor="@android:color/black" />

    <EditText
        android:id="@+id/studentNameEdit"
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:hint="Student Name" android:layout_marginTop="10dp"
        android:textColor="@android:color/black"/>

    <Button
        android:id="@+id/issueBookBtn"
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:text="Issue Book" android:layout_marginTop="16dp" />

    <TextView
        android:id="@+id/statusTextView"
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:text="No books issued currently."
        android:textColor="@android:color/black"
        android:textSize="16sp" android:layout_marginTop="18dp" />
</LinearLayout>
```

Output:



Practical -11

Aim:- Create an activity using Grid to position elements.

```
package com.example.griddemoactivity

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.fillMaxSize
import androidx.compose.material3.MaterialTheme
import androidx.compose.material3.Surface
import androidx.compose.material3.Text
import androidx.compose.runtime.Composable
import androidx.compose.ui.Modifier
import androidx.compose.ui.tooling.preview.Preview
import com.example.griddemoactivity.ui.theme.GridDemoActivityTheme

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContent {
            GridDemoActivityTheme {
                // A surface container using the 'background' color from the theme
                Surface(modifier = Modifier.fillMaxSize(), color =
MaterialTheme.colorScheme.background) {
                    Greeting("Android")
                }
            }
        }
    }
}

@Composable
fun Greeting(name: String, modifier: Modifier = Modifier) {
    Text(
        text = "Hello $name!",
        modifier = modifier
    )
}

@Preview(showBackground = true)
@Composable
fun GreetingPreview() {
    GridDemoActivityTheme {
        Greeting("Android")
```

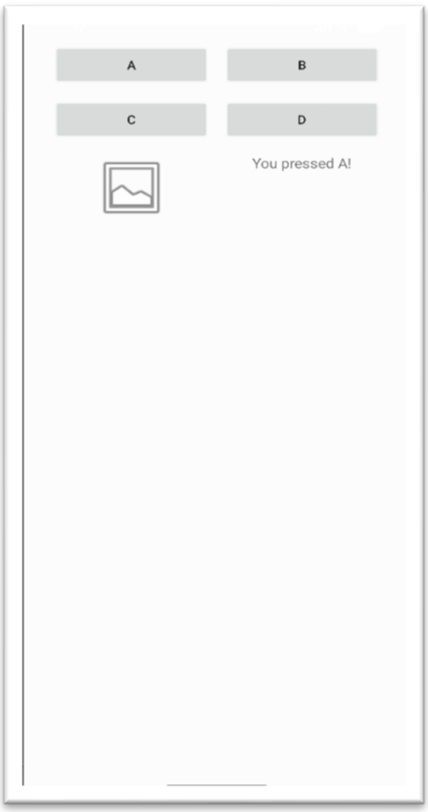
```
    }
}
```

//activity_grid_demo.xml

```
<androidx.gridlayout.widget.GridLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/gridLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="24dp"
    app:columnCount="2"
    app:rowCount="3">

    <Button
        android:id="@+id/button1"
        app:layout_row="0"
        app:layout_column="0"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:text="A"
        app:layout_columnWeight="1"
        android:layout_margin="8dp"/>

    <Button
        android:id="@+id/button2"
        app:layout_row="0"
        app:layout_column="1"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:text="B"
        app:layout_columnWeight="1"
        android:layout_margin="8dp"/>


    <Button
        android:id="@+id/button4"
        app:layout_row="1"
        app:layout_column="1"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:text="D"
        app:layout_columnWeight="1"
        android:layout_margin="8dp"/>
```

```
  <ImageView
      android:id="@+id/imageView"
      app:layout_row="2"
      app:layout_column="0"
      android:layout_width="0dp"
      android:layout_height="80dp"
      android:src="@android:drawable/ic_menu_camera"
      app:layout_columnWeight="1"
      android:layout_margin="8dp"/>

   <TextView
      android:id="@+id/resultText"
      app:layout_row="2"
      app:layout_column="1"
      android:layout_width="0dp"
      android:layout_height="wrap_content"
      android:text="Grid Output will show here"
      android:textSize="16sp"
      android:gravity="center"
      app:layout_columnWeight="1"
      android:layout_margin="8dp"/>

</androidx.gridlayout.widget.GridLayout>
```

Output:



Practical -12

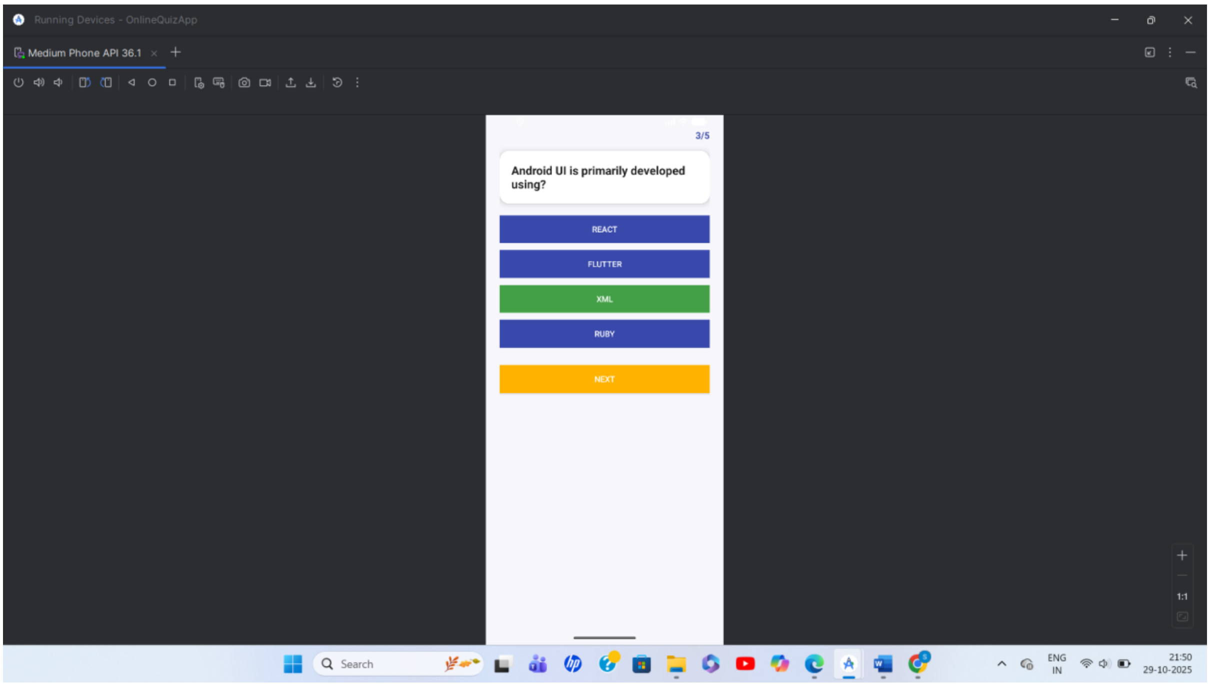
Aim:- Develop an Online quiz application to make quizzes for requirements.

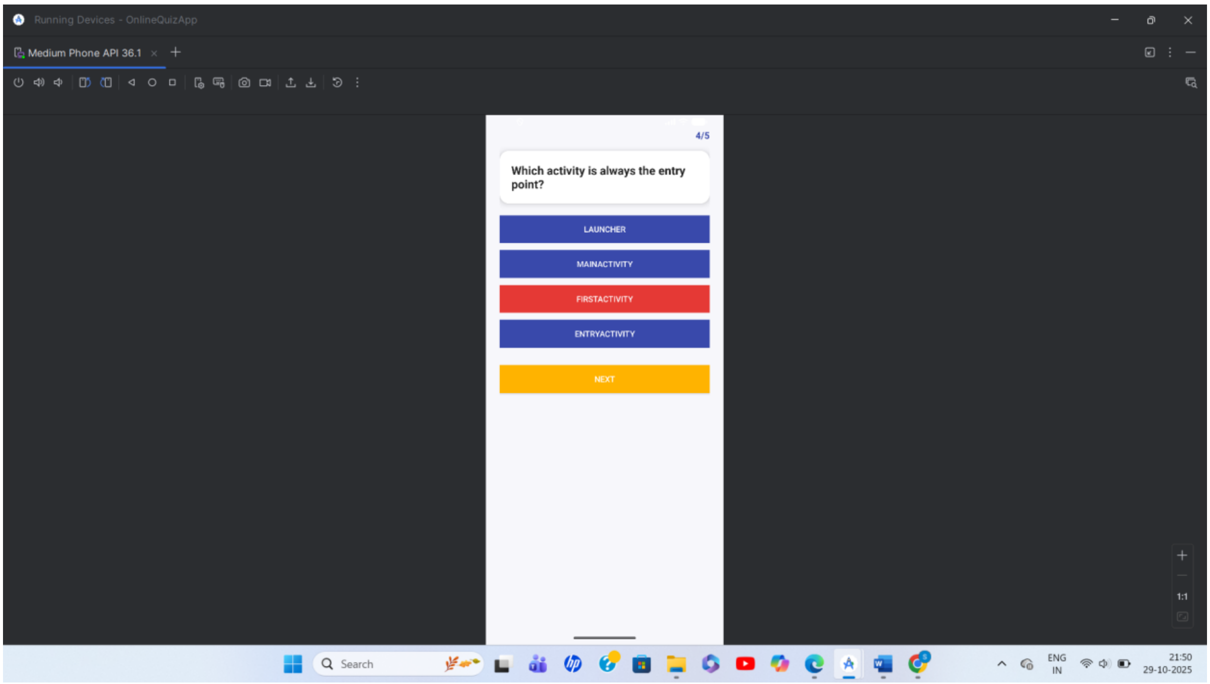
```
package com.example.onlinequizapp

import android.graphics.Color
import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity
import androidx.core.content.ContextCompat

data class Question(
    val text: String,
    val options: List<String>,
    val correct: Int
)

class MainActivity : AppCompatActivity() {
    private val questions = listOf(
        Question("What is Android's official language?", listOf("Swift", "Kotlin", "Dart",
"JavaScript"), 1),
        Question("Who owns Android?", listOf("Microsoft", "Apple", "Google", "IBM"), 2),
        Question("Android UI is primarily developed using?", listOf("React", "Flutter", "XML",
"Ruby"), 2),
        Question("Which activity is always the entry point?", listOf("Launcher",
"MainActivity", "FirstActivity", "EntryActivity"), 1),
        Question("What does SDK stand for?", listOf("Software Development Kit", "Software
Design Kit", "Security Driver Kit", "Service Design Kit"), 0)
    )

    private var qIndex = 0
    private var score = 0

    private lateinit var progress: TextView
    private lateinit var questionText: TextView
    private lateinit var scoreView: TextView
    private lateinit var optionButtons: List<Button>
    private lateinit var nextBtn: Button

    private var selectedIndex = -1

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        progress = findViewById(R.id.questionProgress)
```

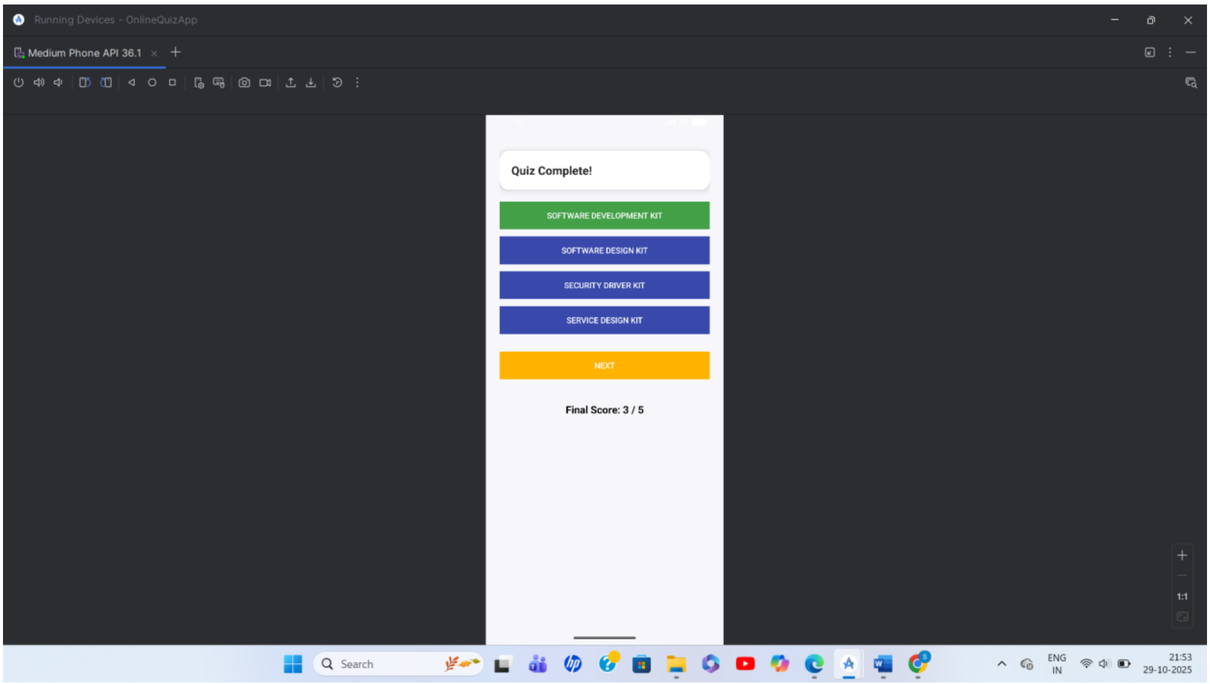
```
    questionText = findViewById(R.id.questionText)
    scoreView = findViewById(R.id.scoreView)
    val optionA: Button = findViewById(R.id.optionA)
    val optionB: Button = findViewById(R.id.optionB)
    val optionC: Button = findViewById(R.id.optionC)
    val optionD: Button = findViewById(R.id.optionD)
    nextBtn = findViewById(R.id.nextBtn)

    optionButtons = listOf(optionA, optionB, optionC, optionD)

    showQuestion()

    optionButtons.forEachIndexed { idx, btn ->
        btn.setOnClickListener {
            if (selectedIndex == -1) {
                selectedIndex = idx
                highlightAnswer(idx)
                nextBtn.isEnabled = true
            }
        }
    }

    nextBtn.setOnClickListener {
        if (selectedIndex == questions[qIndex].correct) score++
        qIndex++
        if (qIndex < questions.size) {
            selectedIndex = -1
            showQuestion()
        } else {
            showResult()
        }
    }
}

private fun highlightAnswer(idx: Int) {
    optionButtons.forEachIndexed { i, btn ->
        btn.setBackgroundColor(ContextCompat.getColor(this, if (i == idx) {
            if (i == questions[qIndex].correct) R.color.correct else R.color.wrong
        } else R.color.primary))
    }
}

private fun showQuestion() {
    val q = questions[qIndex]
    progress.text = "${qIndex + 1}/${questions.size}"
    questionText.text = q.text
```

```
        optionButtons.forEachIndexed { i, b ->
            b.text = q.options[i]
            b.setBackgroundColor(ContextCompat.getColor(this, R.color.primary))
        }
        nextBtn.isEnabled = false
        scoreView.text = ""
    }

    private fun showResult() {
        questionText.text = "Quiz Complete!"
        progress.text = ""
        optionButtons.forEach { it.isEnabled = false }
        nextBtn.isEnabled = false
        scoreView.text = "Final Score: $score / ${questions.size}"
    }
}
```

Output:




Running Devices - OnlineQuizApp
Medium Phone API 36.1
4/5
Which activity is always the entry point?
LAUNCHER
MAINACTIVITY
FIRSTACTIVITY
ENTRYACTIVITY
NEXT
Search
ENG IN
21:50
29-10-2025


Running Devices - OnlineQuizApp
Medium Phone API 36.1
Quiz Complete!
SOFTWARE DEVELOPMENT KIT
SOFTWARE DESIGN KIT
SECURITY DRIVER KIT
SERVICE DESIGN KIT
NEXT
Final Score: 3 / 5
Search
ENG IN
21:53
29-10-2025

Practical -13

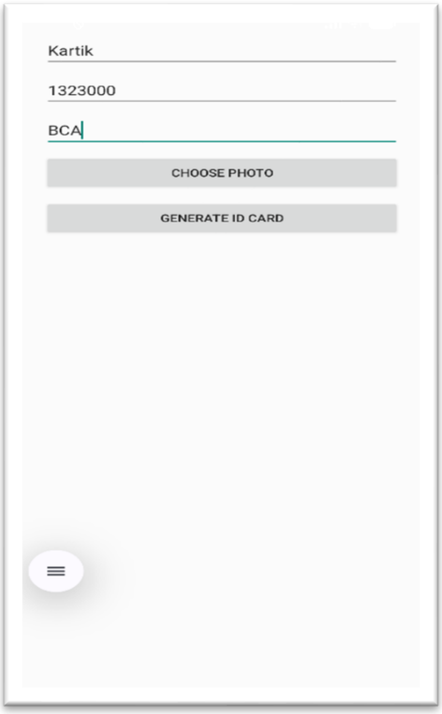
Aim:- Develop an ID Card Generator application for verified students.

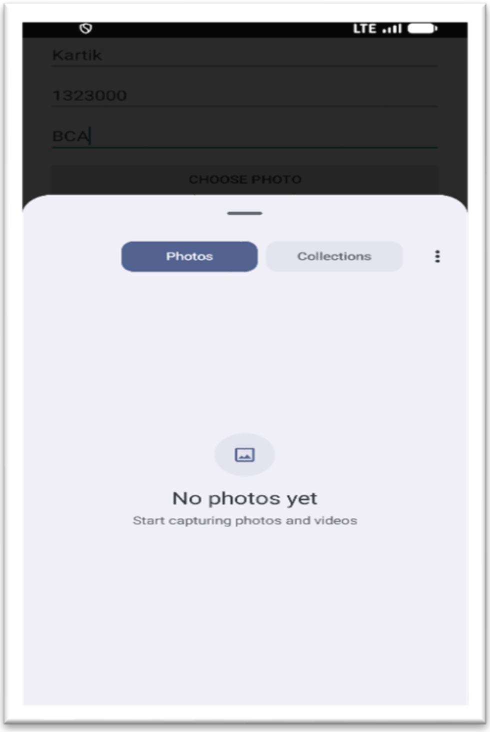
```
package com.example.studentidcardgenerator

import android.Manifest
import android.app.Activity
import android.content.Intent
import android.content.pm.PackageManager
import android.net.Uri
import android.os.Build
import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity
import androidx.core.app.ActivityCompat

class MainActivity : AppCompatActivity() {
   private var photoUri: Uri? = null
   private val PERMISSION_REQUEST_CODE = 200

   override fun onCreate(savedInstanceState: Bundle?) {
      super.onCreate(savedInstanceState)
      setContentView(R.layout.activity_main)

      val nameEdit = findViewById<EditText>(R.id.nameEdit)
      val rollEdit = findViewById<EditText>(R.id.rollEdit)
      val courseEdit = findViewById<EditText>(R.id.courseEdit)
      val choosePhotoBtn = findViewById<Button>(R.id.choosePhotoBtn)
      val generateCardBtn = findViewById<Button>(R.id.generateCardBtn)
      val cardView = findViewById<androidx.cardview.widget.CardView>(R.id.idCardView)
      val cardName = findViewById<TextView>(R.id.cardName)
      val cardRoll = findViewById<TextView>(R.id.cardRoll)
      val cardCourse = findViewById<TextView>(R.id.cardCourse)
      val photoView = findViewById<ImageView>(R.id.photoView)

      choosePhotoBtn.setOnClickListener {
         val permission = if (Build.VERSION.SDK_INT >=
Build.VERSION_CODES.TIRAMISU) {
            Manifest.permission.READ_MEDIA_IMAGES
         } else {
            Manifest.permission.READ_EXTERNAL_STORAGE
         }

         if (checkSelfPermission(permission) == PackageManager.PERMISSION_DENIED) {
            ActivityCompat.requestPermissions(this, arrayOf(permission),
PERMISSION_REQUEST_CODE)
```

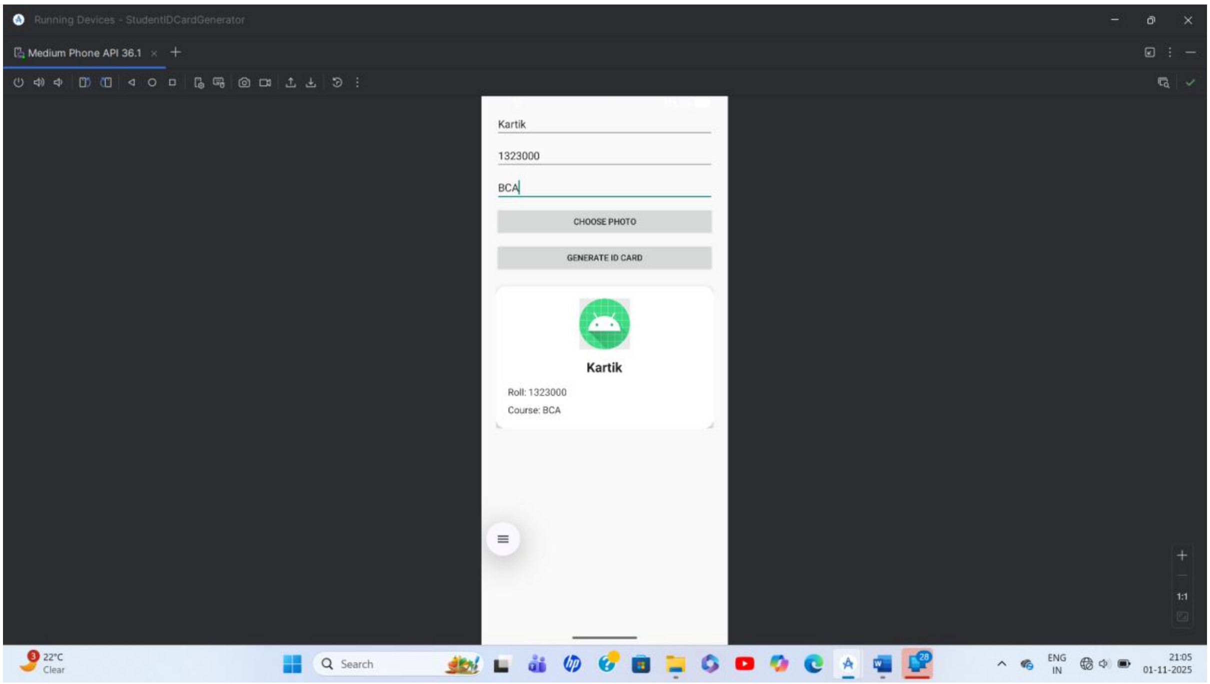
```
            } else {
                openGallery()
            }
        }

        generateCardBtn.setOnClickListener {
            val name = nameEdit.text.toString()
            val roll = rollEdit.text.toString()
            val course = courseEdit.text.toString()
            if (name.isEmpty() || roll.isEmpty() || course.isEmpty()) {
                Toast.makeText(this, "Please fill all fields", Toast.LENGTH_SHORT).show()
            } else {
                cardName.text = name
                cardRoll.text = "Roll: $roll"
                cardCourse.text = "Course: $course"
                if (photoUri != null) photoView.setImageURI(photoUri)
                cardView.visibility = android.view.View.VISIBLE
            }
        }
    }

    private fun openGallery() {
        val pickIntent = Intent(Intent.ACTION_GET_CONTENT)
        pickIntent.type = "image/*"
        startActivityForResult(pickIntent, 100)
    }

    override fun onActivityResult(requestCode: Int, resultCode: Int, data: Intent?) {
        super.onActivityResult(requestCode, resultCode, data)
        if (requestCode == 100 && resultCode == Activity.RESULT_OK) {
            photoUri = data?.data
        }
    }

    override fun onRequestPermissionsResult(requestCode: Int, permissions: Array<out
String>, grantResults: IntArray) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults)
        if (requestCode == PERMISSION_REQUEST_CODE) {
            if (grantResults.isNotEmpty() && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
                openGallery()
            } else {
                Toast.makeText(this, "Permission Denied", Toast.LENGTH_SHORT).show()
            }
        }
```

```
    }
}
```

Output:


Kartik
1323000
BCA
CHOOSE PHOTO
GENERATE ID CARD


LTE
Kartik
1323000
BCA
CHOOSE PHOTO
Photos
Collections
No photos yet
Start capturing photos and videos


Running Devices - StudentIDCardGenerator
Medium Phone API 36.1
Kartik
1323000
BCA
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Practical -14

Aim:- Create an app using Alarm Manager and Content Provider.

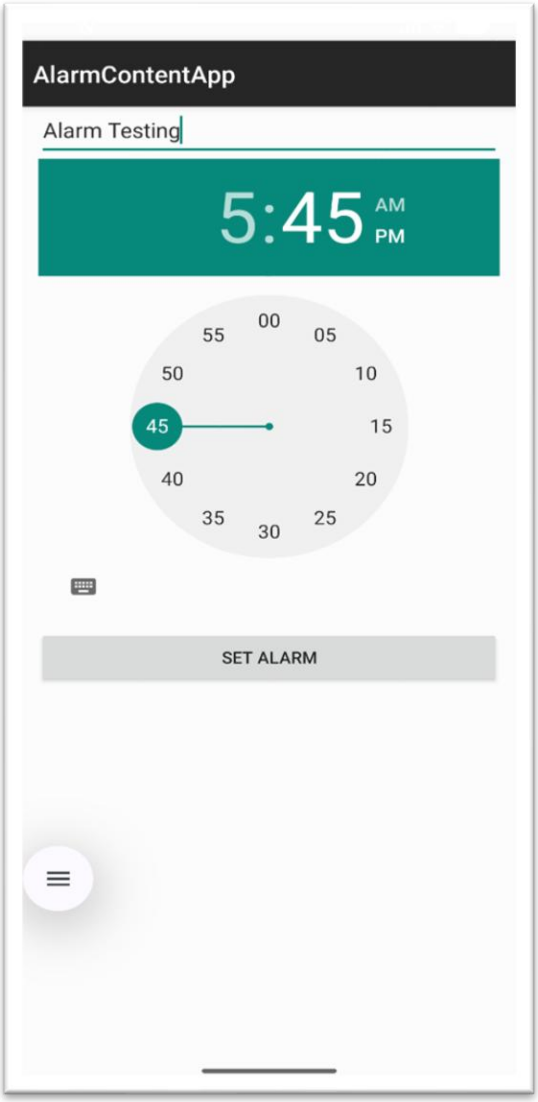
```
package com.example.alarmcontentapp

import android.app.AlarmManager
import android.app.PendingIntent
import android.content.Context
import android.content.Intent
import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity
import java.util.*

class MainActivity : AppCompatActivity() {

    private lateinit var titleInput: EditText
    private lateinit var timePicker: TimePicker
    private lateinit var setBtn: Button
    private lateinit var statusText: TextView

    private val prefName = "AlarmPrefs"
    private val keyAlarmTitle = "alarm_title"
    private val keyAlarmTime = "alarm_time"

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        titleInput = findViewById(R.id.titleInput)
        timePicker = findViewById(R.id.timePicker)
        setBtn = findViewById(R.id.setBtn)
        statusText = findViewById(R.id.statusText)

        // The line below was causing a persistent, incorrect compiler error and has been
removed.
        // timePicker.is24HourView = true

        setBtn.setOnClickListener {
            val title = titleInput.text.toString().ifEmpty { "Alarm" }
            val hour = timePicker.hour
            val minute = timePicker.minute

            val calendar = Calendar.getInstance().apply {
                set(Calendar.HOUR_OF_DAY, hour)
                set(Calendar.MINUTE, minute)
```

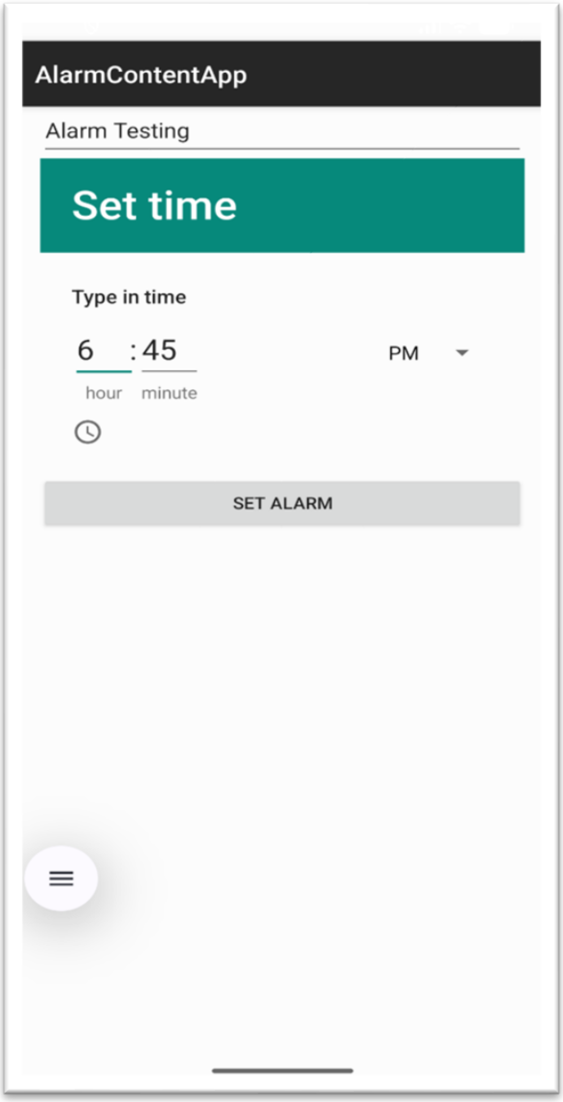
```
                set(Calendar.SECOND, 0)
                if (before(Calendar.getInstance())) add(Calendar.DATE, 1)
            }

            // Save to SharedPreferences
            val prefs = getSharedPreferences(prefName, Context.MODE_PRIVATE)
            with(prefs.edit()) {
                putString(keyAlarmTitle, title)
                putLong(keyAlarmTime, calendar.timeInMillis)
                apply()
            }

            // Schedule alarm
            val intent = Intent(this, AlarmReceiver::class.java).apply {
                putExtra("title", title)
            }
            val pi = PendingIntent.getBroadcast(this, 0, intent,
PendingIntent.FLAG_UPDATE_CURRENT or PendingIntent.FLAG_IMMUTABLE)
            val alarmManager = getSystemService(Context.ALARM_SERVICE) as
AlarmManager
            alarmManager.setExact(AlarmManager.RTC_WAKEUP, calendar.timeInMillis, pi)

            statusText.text = "Alarm set for $hour:$minute - $title"
        }
    }
}
```

```
//activity_main.xml

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="20dp">

    <EditText
        android:id="@+id/titleInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Alarm Title"
        android:minHeight="48dp" />

    <TimePicker
        android:id="@+id/timePicker"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />

    <Button
        android:id="@+id/setBtn"
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Set Alarm"
        android:layout_marginTop="12dp"/>

    <TextView
        android:id="@+id/statusText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:textSize="16sp"/>
</LinearLayout>
```

Output:





Practical -15

Aim:- Develop a simple login and registration form using Java and Android Studio.

```
package com.example.loginregisterapp;

import android.content.Intent;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class LoginActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_login);

        EditText userEdit = findViewById(R.id.userEdit);
        EditText passEdit = findViewById(R.id.passEdit);
        Button loginBtn = findViewById(R.id.loginBtn);
        Button gotoRegisterBtn = findViewById(R.id.gotoRegisterBtn);
        TextView statusText = findViewById(R.id.statusText);

        loginBtn.setOnClickListener(v -> {
            String user = userEdit.getText().toString();
            String pass = passEdit.getText().toString();

            if(user.equals(RegisterActivity.registeredUser) &&
pass.equals(RegisterActivity.registeredPass)){
                statusText.setText("Login successful!");
            } else {
                statusText.setText("Invalid username or password.");
            }
        });

        gotoRegisterBtn.setOnClickListener(v -> {
            startActivity(new Intent(this, RegisterActivity.class));
            finish();
        });
    }
}
```

```
//RegisterActivity.java

package com.example.loginregisterapp;
```

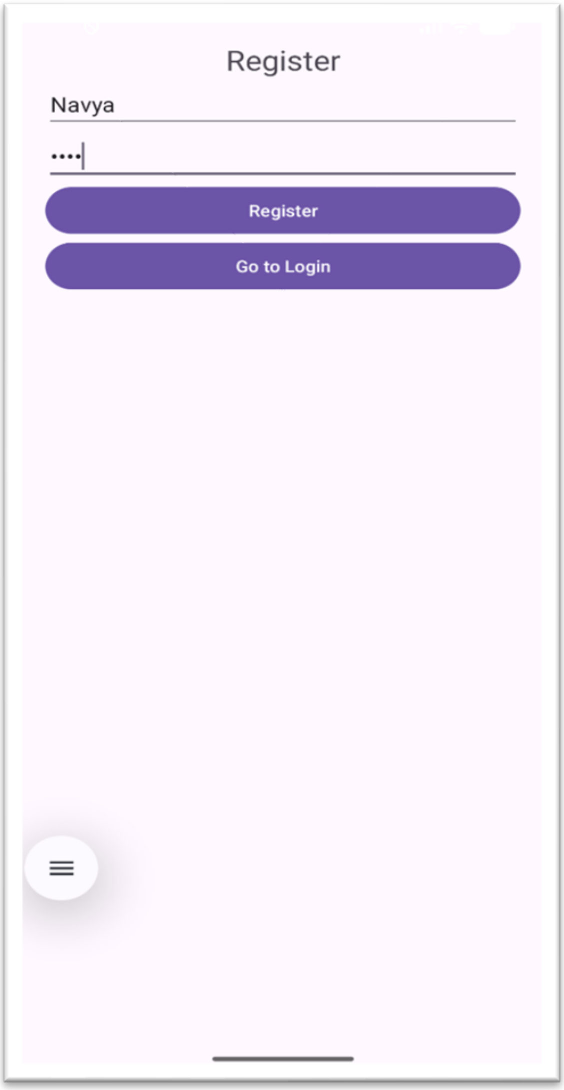
```
import android.content.Intent;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class RegisterActivity extends AppCompatActivity {

    public static String registeredUser = "";
    public static String registeredPass = "";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_register);

        EditText userEdit = findViewById(R.id.userEdit);
        EditText passEdit = findViewById(R.id.passEdit);
        Button registerBtn = findViewById(R.id.registerBtn);
        Button gotoLoginBtn = findViewById(R.id.gotoLoginBtn);

        registerBtn.setOnClickListener(v -> {
            registeredUser = userEdit.getText().toString();
            registeredPass = passEdit.getText().toString();

            if(registeredUser.isEmpty() || registeredPass.isEmpty()) {
                Toast.makeText(this, "Enter both username and password",
Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(this, "Registration successful! Please login.",
Toast.LENGTH_SHORT).show();
            }
        });

        gotoLoginBtn.setOnClickListener(v -> {
            startActivity(new Intent(this, LoginActivity.class));
            finish();
        });
    }
}
```

//activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
```

```
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



Practical -16

Aim:- Build a chat application to interact with people around the world.

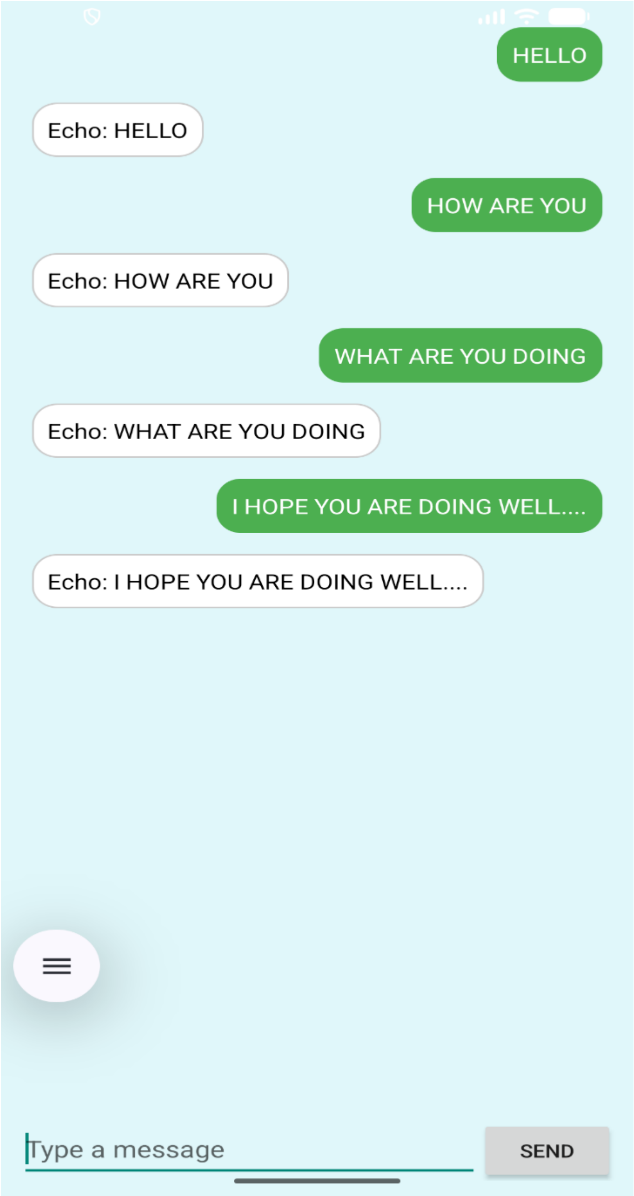
```
package com.example.bubblechatapp

import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import androidx.appcompat.app.AppCompatActivity
import androidx.recyclerview.widget.LinearLayoutManager
import androidx.recyclerview.widget.RecyclerView

class MainActivity : AppCompatActivity() {

    private lateinit var chatRecyclerView: RecyclerView
    private lateinit var messageInput: EditText
    private lateinit var sendButton: Button
    private lateinit var adapter: ChatAdapter
    private val messageList = mutableListOf<Message>()

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        chatRecyclerView = findViewById(R.id.chatRecyclerView)
        messageInput = findViewById(R.id.messageInput)
        sendButton = findViewById(R.id.sendButton)

        adapter = ChatAdapter(messageList)
        chatRecyclerView.layoutManager = LinearLayoutManager(this)
        chatRecyclerView.adapter = adapter

        sendButton.setOnClickListener {
            val text = messageInput.text.toString()
            if (text.isNotEmpty()) {
                addMessage(text, true)
                messageInput.text.clear()

                // Simulate received message (replace with actual network code)
                chatRecyclerView.postDelayed({ addMessage("Echo: $text", false) }, 1000)
            }
        }
    }

    private fun addMessage(text: String, isSent: Boolean) {
        messageList.add(Message(text, isSent))
```

```
        adapter.notifyItemInserted(messageList.size - 1)
        chatRecyclerView.scrollToPosition(messageList.size - 1)
    }
}
```

//activity_main.xml

```xml
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:orientation="vertical" android:padding="12dp"
    android:background="#E0F7FA">

    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/chatRecyclerView"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:paddingBottom="8dp" />

    <LinearLayout
        android:layout_width="match_parent" android:layout_height="wrap_content"
        android:orientation="horizontal">

        <EditText
            android:id="@+id/messageInput"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:hint="Type a message"
            android:minHeight="48dp" />

        <Button
            android:id="@+id/sendButton"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:minHeight="48dp"
            android:text="Send" />
    </LinearLayout>
</LinearLayout>
```

Output:


HELLO
Echo: HELLO
HOW ARE YOU
Echo: HOW ARE YOU
WHAT ARE YOU DOING
Echo: WHAT ARE YOU DOING
I HOPE YOU ARE DOING WELL....
Echo: I HOPE YOU ARE DOING WELL....
Type a message
SEND